

Input Form (IF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: CRISISLTLSUM | Context: Ecuador — Humanitarian Crisis Social Media Analysts

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's preprocessing pipeline is English-specific throughout: AllenNLP NER trained on CoNLL03 [Q33], BERT-based NER fine-tuned on English Twitter [Q35], lemmatization and lowercasing of English keywords [Q129], OpenStreetMap matching of English place names [Q36], and BERT/BART models with 512-token English-tokenizer constraints [Q86, Q141, Q143]. The authors themselves flag that 'replicating the same process for other language or based on other locations may be hard' [Q119]. The Ecuador deployment processes Ecuadorian Spanish in informal/semi-formal registers with Kichwa-Spanish code-switching and crisis-specific vocabulary (elicitation A4); no Kichwa NLP resources are publicly available [WEB-1, WEB-13]. The signal-distribution mismatch is total at the language level and severe at the register/script level — the benchmark cannot process Spanish input at all without reconstruction, and Kichwa toponyms within Spanish tweets would not be recognized by any component of the pipeline.

ID	Evidence
Q33	"We use three different modules. First, we use a pre-trained neural model from AllenNLP (Gardner et al., 2018), trained on CoNLL03 (Tjong Kim Sang and De Meulder, 2003), to extract entities with types of people, location, and organization." (p.3)
Q35	"Since location mentions are crucial in extracting local events and existing models have low performance detecting them from noisy tweets text, we further developed a BERT-based NER model tuned on Twitter data to detect location mentions." (p.4)
Q36	"Location Augmentation We use OpenStreetMap API to map location mentions to physical addresses." (p.4)
Q86	"For our first sentence-level classification model, we fine-tune BERT (Devlin et al., 2019) on the tweet sequences where every training example would have the form of "t0 [SEP] si"; where si ∈ S." (p.7)
Q119	"Accordingly, replicating the same process for other language or based on other locations may be hard because of such dependencies." (p.10)
Q129	"To avoid making a long list of keywords and ensure that different lexical formats of the same word are still considered in the keyword matching process, we maintain both a lower-cased and a lemmatized version of the keywords and the tweet's text." (p.13)
Q141	"We fine-tune BERT base uncased on a single GPU for 10 epochs with a learning rate of 5e-5, batch size of 32, a seed of 42, and a maximum sequence length of 512." (p.14)
Q143	"We fine-tune BART based on a single GPU for ten epochs with a learning rate of 5e-5, batch size of 16, a seed of 42, and a maximum target sequence length of 512." (p.14)
WEB-1	Kichwa orthography is contested with continued spelling variation; written use of indigenous languages outside schools is minimal. (https://books.openbookpublishers.com/10.11647/obp.0032/chapter11.html)
WEB-13	First Ecuadorian Spanish NLP corpus exists but does not cover Kichwa. (https://ieeexplore.ieee.org/document/7889589)

Strengths

- The benchmark is text-only and Twitter-native, matching the surface modality of the Ecuadorian deployment's primary data source; UrbangEnCy [WEB-17] confirms Twitter is also used for Ecuadorian emergency NLP, so the platform-level signal type is shared.

ID	Evidence
WEB-17	UrbangEnCy demonstrates feasibility of Spanish Ecuadorian Twitter capture but is binary-classification only. (https://www.sciencedirect.com/science/article/pii/S2352340920315729)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distribution mismatch is severe: the benchmark operates on English Twitter; the deployment requires Ecuadorian Spanish with code-switching, indigenous toponyms, and informal orthography [WEB-1]. Tokenizers, NER, lemmatization [Q129], and place-name matching [Q36] would all need replacement.

IF-2: Regional infrastructure supports Twitter capture (95.8% mobile broadband [WEB-15]; UrbangEnCy demonstrates feasibility [WEB-17]), but Amazonian rural areas have severely limited connectivity, biasing the available signal away from indigenous communities most exposed to environmental crises.

IF-3: Domain-specific form differences include Kichwa orthographic variation (the 2008 unified alphabet remains contested [WEB-1]), Spanish accent marks, and crisis vocabulary like 'damnificados', 'lahares' that no English-trained tokenizer or NER will handle correctly.

IF-4: External validity is fundamentally violated: the entire preprocessing pipeline [Q28–Q37] is English/US/Australia-calibrated and the authors explicitly acknowledge the language dependency [Q119].

ID	Evidence
Q28	"Figure 2 shows the process for generating a set of noisy timelines starting from the Twitter stream and followed by pre-processing and knowledge enhancement steps, the online clustering method, and post-processing & cleaning steps." (p.3)
Q29	"We limit the incoming tweets to specific geographical areas, periods, and domains of interest." (p.3)
Q30	"The location filtering relies on a list of location candidates created by gathering cities, towns, and famous neighborhoods in a big area of interest. A tweet is considered relevant to our area of interest if 1) the text mentions one of the candidates, 2) the geo-tag matches the area of interest, or 3) the user location matches the area of interest." (p.3)
Q31	"To limit the tweets to a specified crisis domain, we curate domain-specific keywords and only select tweets with phrases matching one of the keywords. This approach is not comprehensive or exhaustive but somewhat representative of each crisis domain." (p.3)
Q32	"The combinations of (area a, domain d, time t) are manually selected so that the events of type d at location a are more frequent during time period t." (p.3)
Q33	"We use three different modules. First, we use a pre-trained neural model from AllenNLP (Gardner et al., 2018), trained on CoNLL03 (Tjong Kim Sang and De Meulder, 2003), to extract entities with types of people, location, and organization." (p.3)
Q34	"Although this module extracts some important entities in the text, it fails to extract uncommon entities or special mentions" (p.3)
Q35	"Since location mentions are crucial in extracting local events and existing models have low performance detecting them from noisy tweets text, we further developed a BERT-based NER model tuned on Twitter data to detect location mentions." (p.4)
Q36	"Location Augmentation We use OpenStreetMap API to map location mentions to physical addresses." (p.4)
Q37	"This step provides complementary information about each location while reducing the noise introduced by the entity extraction module through removing location mentions that are wrongly detected or are not located in the area of interest." (p.4)
Q119	"Accordingly, replicating the same process for other language or based on other locations may be hard because of such dependencies." (p.10)
Q129	"To avoid making a long list of keywords and ensure that different lexical formats of the same word are still considered in the keyword matching process, we maintain both a lower-cased and a lemmitized version of the keywords and the tweet's text." (p.13)
WEB-1	Kichwa orthography is contested with continued spelling variation; written use of indigenous languages outside schools is minimal. (https://books.openbookpublishers.com/10.11647/obp.0032/chapter11.html)
WEB-15	Ecuador has 95.8% mobile broadband connections, supporting Twitter use. (https://datareportal.com/reports/digital-2026-ecuador)

WEB-17	UrbangEnCy demonstrates feasibility of Spanish Ecuadorian Twitter capture but is binary-classification only. (https://www.sciencedirect.com/science/article/pii/S2352340920315729)
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Information Gaps

- OSM coverage quality for Ecuadorian rural geography; smartphone vs. feature-phone distribution in Oriente.

Requires Expert Verification

- Frequency and patterns of Kichwa-Spanish code-switching in real-time crisis tweets from indigenous federation accounts.

Remediation

Gap 1: English-only preprocessing pipeline (NER, lemmatization, OSM, tokenizers) cannot process Spanish or Kichwa.

Recommendation: Replace AllenNLP/CoNLL03 NER with Spanish NER fine-tuned on Ecuadorian tweets (extending the Mexico earthquake template [WEB-20]); validate OpenStreetMap coverage for Ecuadorian rural geography and supplement with a Kichwa-toponym gazetteer; handle the SGR/SNGR acronym alias [WEB-11] explicitly.

ID	Evidence
WEB-11	SNGR was formally renamed from SGR in 2024; both acronyms appear in tweets. (https://www.gestionderiesgos.gob.ec/wp-content/uploads/downloads/2024/05/Resol.SNGR-102-2024.pdf)
WEB-20	https://pmc.ncbi.nlm.nih.gov/articles/PMC6484392/